

Problem Solving Using Python (CSE1012)

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Factorial Computation

- Example,
- $0! = 1$
- $1! = 1$
- $2! = 2 \times 1 = 2$
- $5! = 5 \times 4 \times 3 \times 2 \times 1 = 120$

Factorial Computation

Factorial Computation

```
step 1. Start
step 2. Read the number n
step 3. [Initialize]
        i=1, fact=1
step 4. Repeat step 4 through 6 until i=n
step 5. fact=fact*i
step 6. i=i+1
step 7. Print fact
step 8. Stop
```

Factorial Computation

- $\text{fact} = 1$
- Set a number N for which factorial to be calculated
- IF $N > 1$
 - REPEAT FOR $i = 1$ to N
 - $\text{fact} = \text{fact} * i$
 - END FOR
- END IF
- Print fact

Sine Function Computation

- Write an algorithm to compute $\text{sine}(x)$ for given x
- Supply x and a positive integer n .
- Compute $\text{sine}(x)$ using the series and the computation should use all terms in the series up through the term involving x^n

$$\text{sine } x = x - \left(\frac{x^3}{3!}\right) + \left(\frac{x^5}{5!}\right) - \left(\frac{x^7}{7!}\right) + \dots$$

Sine Function Computation

$$\begin{aligned}\sin x &= x - \frac{x^3}{3!} + \frac{x^5}{5!} - \frac{x^7}{7!} + \cdots \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n+1}}{(2n+1)!}, \\ \cos x &= 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \frac{x^6}{6!} + \cdots \\ &= \sum_{n=0}^{\infty} \frac{(-1)^n x^{2n}}{(2n)!}.\end{aligned}$$

Fibonacci Sequence

- Each number is the sum of previous two numbers
- 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144,...
- How to generate the series of numbers?

Fibonacci Sequence

- We need: 0, 1, 1, 2, 3, 5, 8, 13, 21, 34, 55, 89, 144...
- Set $F(0) = 0$, and $F(1) = 1$
- Generate $F(n)$ for $n > 1$ as,

$$F_n = F_{n-1} + F_{n-2}$$

Reversing the digits of an Integer

- Example,
- 513 -> 315

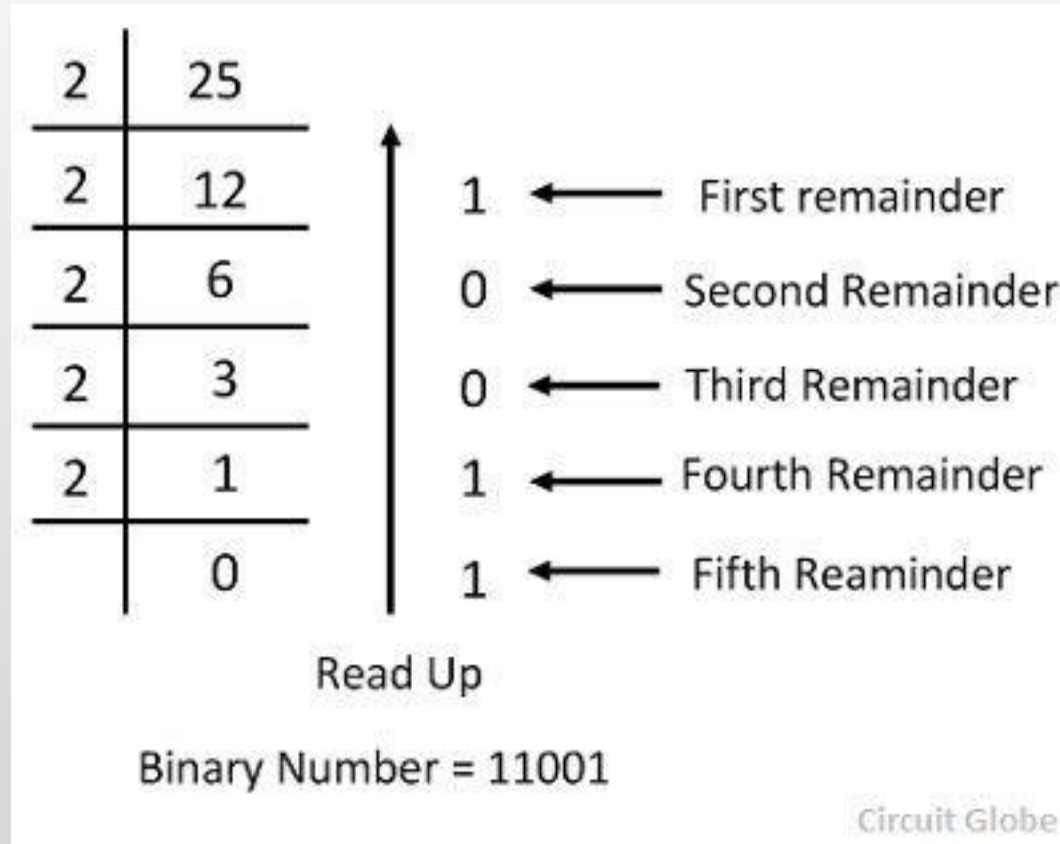
Reversing the digits of an Integer

- Input a number I
- $Rev = 0$
- While $I > 0$
 - $Rev = Rev * 10 + I \% 10$
 - $I = I // 10$
- Print Rev

Base Conversion

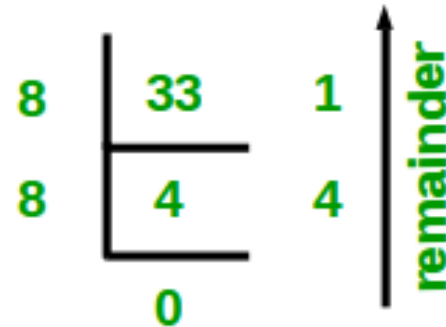
- Decimal to binary
- Decimal to octal
- Decimal to Hexadecimal

Decimal to binary



Decimal to octal

Decimal Number: 33



Octal Number: 41

Decimal to Hexadecimal

4735_{10}

$$4735/16 = 295$$

$$295/16 = 18$$

$$18/16 = 1$$

$$1/16 = 0$$

$$15 = F$$

$$7 = 7$$

$$2 = 2$$

$$1 = 1$$

$127F_{16}$

$$10 = A$$

$$11 = B$$

$$12 = C$$

$$13 = D$$

$$14 = E$$

$$15 = F$$

ASCII Conversion Chart

Decimal - Binary - Octal - Hex – ASCII Conversion Chart																			
Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII	Decimal	Binary	Octal	Hex	ASCII
0	00000000	000	00	NUL	32	00100000	040	20	SP	64	01000000	100	40	@	96	01100000	140	60	`
1	00000001	001	01	SOH	33	00100001	041	21	!	65	01000001	101	41	A	97	01100001	141	61	a
2	00000010	002	02	STX	34	00100010	042	22	"	66	01000010	102	42	B	98	01100010	142	62	b
3	00000011	003	03	ETX	35	00100011	043	23	#	67	01000011	103	43	C	99	01100011	143	63	c
4	00000100	004	04	EOT	36	00100100	044	24	\$	68	01000100	104	44	D	100	01100100	144	64	d
5	00000101	005	05	ENQ	37	00100101	045	25	%	69	01000101	105	45	E	101	01100101	145	65	e
6	00000110	006	06	ACK	38	00100110	046	26	&	70	01000110	106	46	F	102	01100110	146	66	f
7	00000111	007	07	BEL	39	00100111	047	27	'	71	01000111	107	47	G	103	01100111	147	67	g
8	00001000	010	08	BS	40	00101000	050	28	(	72	01001000	110	48	H	104	01101000	150	68	h
9	00001001	011	09	HT	41	00101001	051	29	)	73	01001001	111	49	I	105	01101001	151	69	i
10	00001010	012	0A	LF	42	00101010	052	2A	*	74	01001010	112	4A	J	106	01101010	152	6A	j
11	00001011	013	0B	VT	43	00101011	053	2B	+	75	01001011	113	4B	K	107	01101011	153	6B	k
12	00001100	014	0C	FF	44	00101100	054	2C	,	76	01001100	114	4C	L	108	01101100	154	6C	l
13	00001101	015	0D	CR	45	00101101	055	2D	-	77	01001101	115	4D	M	109	01101101	155	6D	m
14	00001110	016	0E	SO	46	00101110	056	2E	.	78	01001110	116	4E	N	110	01101110	156	6E	n
15	00001111	017	0F	SI	47	00101111	057	2F	/	79	01001111	117	4F	O	111	01101111	157	6F	o
16	00010000	020	10	DLE	48	00110000	060	30	0	80	01010000	120	50	P	112	01110000	160	70	p
17	00010001	021	11	DC1	49	00110001	061	31	1	81	01010001	121	51	Q	113	01110001	161	71	q
18	00010010	022	12	DC2	50	00110010	062	32	2	82	01010010	122	52	R	114	01110010	162	72	r
19	00010011	023	13	DC3	51	00110011	063	33	3	83	01010011	123	53	S	115	01110011	163	73	s
20	00010100	024	14	DC4	52	00110100	064	34	4	84	01010100	124	54	T	116	01110100	164	74	t
21	00010101	025	15	NAK	53	00110101	065	35	5	85	01010101	125	55	U	117	01110101	165	75	u
22	00010110	026	16	SYN	54	00110110	066	36	6	86	01010110	126	56	V	118	01110110	166	76	v
23	00010111	027	17	ETB	55	00110111	067	37	7	87	01010111	127	57	W	119	01110111	167	77	w
24	00011000	030	18	CAN	56	00111000	070	38	8	88	01011000	130	58	X	120	01111000	170	78	x
25	00011001	031	19	EM	57	00111001	071	39	9	89	01011001	131	59	Y	121	01111001	171	79	y
26	00011010	032	1A	SUB	58	00111010	072	3A	:	90	01011010	132	5A	Z	122	01111010	172	7A	z
27	00011011	033	1B	ESC	59	00111011	073	3B	;	91	01011011	133	5B	[	123	01111011	173	7B	{
28	00011100	034	1C	FS	60	00111100	074	3C	<	92	01011100	134	5C	\	124	01111100	174	7C	
29	00011101	035	1D	GS	61	00111101	075	3D	=	93	01011101	135	5D	]	125	01111101	175	7D	}
30	00011110	036	1E	RS	62	00111110	076	3E	>	94	01011110	136	5E	^	126	01111110	176	7E	~
31	00011111	037	1F	US	63	00111111	077	3F	?	95	01011111	137	5F	_	127	01111111	177	7F	DEL

Character to Number Conversion

- #ASCII Character to number conversion
- `char_data = "A"`
- `print(ord(char_data))`

Home Works

- Write the algorithms and/or flowchart of the following:
 - Sine Function Computation
 - Generation of Fibonacci series
 - Convert ASCII code to Number
 - Base conversion (Decimal to Binary and vice-versa)
 - Base conversion (Decimal to Octal and vice-versa)
 - Base conversion (Decimal to Hexadecimal and vice-versa)

Thank You

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